

Deep Learning Latency Attacks and Defenses: A Survey from Object Detection to Large Language and Vision-Language Models

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Adversarial machine learning has historically focused on *integrity*: causing models to produce wrong outputs. A complementary and increasingly consequential axis is *availability*, where the attacker’s objective is not to corrupt a prediction but to make the system fail to respond *in time*. *Latency attacks*—also called energy-latency, slowdown, sponge, or resource-consumption attacks—deliberately inflate the inference-time computation, energy, or wall-clock latency of a deep model, so that a real-time consumer (a vehicle controller, an interactive service, a battery-powered sensor) misses its deadline or exhausts its resources even when the underlying predictions remain nominally accurate. This survey synthesizes a literature that has, until now, been fragmented across distinct communities: physical-world attacks on autonomous-driving perception, sponge examples on convolutional accelerators, slowdown attacks on input-adaptive (multi-exit) networks, efficiency-degradation attacks on neural machine translation and image captioning, and—most recently—output-length inflation, verbose-image, and “infinite-thinking” denial-of-service attacks against large language models (LLMs), vision-language models (VLMs), mixture-of-experts (MoE) routers, and chain-of-thought reasoning models. We organize this body of work along two orthogonal axes: the *exploited bottleneck* (post-processing, temporal state, sensor/cooperative fusion, hardware, dynamic-computation control flow, auto-regressive generation, and training-time poisoning) and the *model class* affected (CNN detectors, dynamic/adaptive networks, transformers, LLMs, and multimodal generators). We present a unified threat-model taxonomy, consolidated quantitative comparisons of reported latency, energy, and amplification factors, and a structured review of the nascent defense landscape—budget enforcement, robust/adaptive training, input transformation, architectural sparsity, runtime monitoring, and system-level serving control. We close with a discussion of cross-cutting open challenges: the absence of standardized benchmarks reporting both timing *and* downstream safety impact, the gap between digital and physically realizable attacks, the new attack surface created by inference-time compute scaling, and the tension between efficiency and robustness that pervades every proposed defense.

Additional Key Words and Phrases: availability attacks, energy-latency attacks, sponge examples, inference latency, denial of service, object detection, non-maximum suppression, multi-exit networks, large language models, vision-language models, mixture of experts, real-time systems, adversarial machine learning

1 Introduction

Deep neural networks are now deployed in settings where *when* a prediction arrives is as important as *whether* it is correct. An autonomous vehicle that perceives a pedestrian one second late has effectively not perceived it at all; an interactive assistant that takes thirty seconds to begin responding is unusable; a battery-powered sensor whose model suddenly draws ten times its budgeted energy will exhaust itself before its mission completes. These deployments expose an attack surface that classical adversarial machine learning largely ignored.

Most adversarial-example research targets *integrity*: an attacker perturbs an input so the model misclassifies it [68]. *Availability* attacks pursue a different goal. Rather than changing the answer, the attacker inflates the cost of producing it—measured in floating-point operations, energy, or latency—so the system cannot respond “within a reasonable time” [6]. We use *latency attack* as an umbrella term for this family, which also appears in the literature under the names energy-latency attack, sponge attack, slowdown attack, efficiency-degradation attack, and resource-consumption denial of service. The unifying mechanism is the deliberate steering of a model, or its host system, toward worst-case computational behavior.

1.1 Why latency attacks matter now

Three trends have converged to make latency attacks a first-class concern. First, *real-time autonomy* couples perception latency directly to physical safety: a delayed detection cascades into delayed planning and control [49]. Second, the field has embraced *input-adaptive efficiency*—early-exit networks, token pruning, skimming transformers, and mixture-of-experts routing—which trade a fixed worst-case cost for a lower *average* cost. This is precisely the property an availability attacker can revoke: by forcing every input onto the slow path, the attacker erases the advertised savings and may even exceed the cost of the static model [11, 28]. Third, the rise of *auto-regressive generation* in LLMs and VLMs means that a small input can induce an arbitrarily large output, and the per-query compute cost is now an attacker-controllable variable [14, 23]. The recent move to inference-time “thinking” in reasoning models multiplies this leverage: a single innocuous-looking prompt can trigger tens of thousands of reasoning tokens [45].

1.2 Scope and contributions

Prior surveys treat pieces of this landscape in isolation. The closest, by Brachemi Meftah et al. [4], covers energy-latency attacks on regular, input-adaptive, and decoder-based models, but predates and only shallowly covers the LLM/VLM era—reasoning-model denial of service, mixture-of-experts routing attacks, and multimodal verbose attacks. The autonomous-driving security literature, meanwhile, has produced a rich set of *physically realizable* latency attacks that are rarely discussed alongside the general deep-learning work. This survey deliberately bridges both. We make the following contributions:

- **A unified taxonomy** (Figure 1) that organizes latency attacks by exploited bottleneck and by affected model class, spanning object detection, dynamic networks, transformers, LLMs, and multimodal generators.
- **A unifying principle**—*intermediate-work amplification* (Section 2.3, Figure 2)—that explains every attack class through a single mechanism and motivates a cross-domain *work-budget* defense abstraction.
- **A threat-model framework** (Section 3) that classifies the attacker interface, knowledge, and targeted pipeline stage, making otherwise incomparable works commensurable.
- **Consolidated quantitative comparisons** (Tables 3, 4, 5) of reported latency, energy, FLOP, and output-length amplification factors, with explicit notes on metric incompatibility.
- **A structured defense review** (Section 10) organized by control mechanism rather than by target, exposing which point of the compute pathway each defense actually bounds.
- **A cross-domain synthesis** (Section 11) showing why perception-side and serving-side defenses do not transfer, plus a review of fragmented evaluation practice (Section 12).
- **A forward-looking analysis** (Sections 13–14) of emerging attack surfaces, including NMS-free detectors, and the open challenges that define the next phase of this research.

2 Background and Definitions

2.1 Integrity versus availability

We adopt the NIST taxonomy of adversarial machine learning, which distinguishes integrity, availability, and confidentiality violations [68]. Integrity attacks aim to change the model’s output; availability attacks aim to degrade its usefulness or serviceability. Latency attacks are availability attacks whose specific lever is computational cost. Crucially, a latency attack may leave the prediction *unchanged*—this is a feature, not a side effect, because output-quality monitors (BLEU for translation, mAP for detection, perplexity for text) will not flag an attack that only changes timing [9, 61].

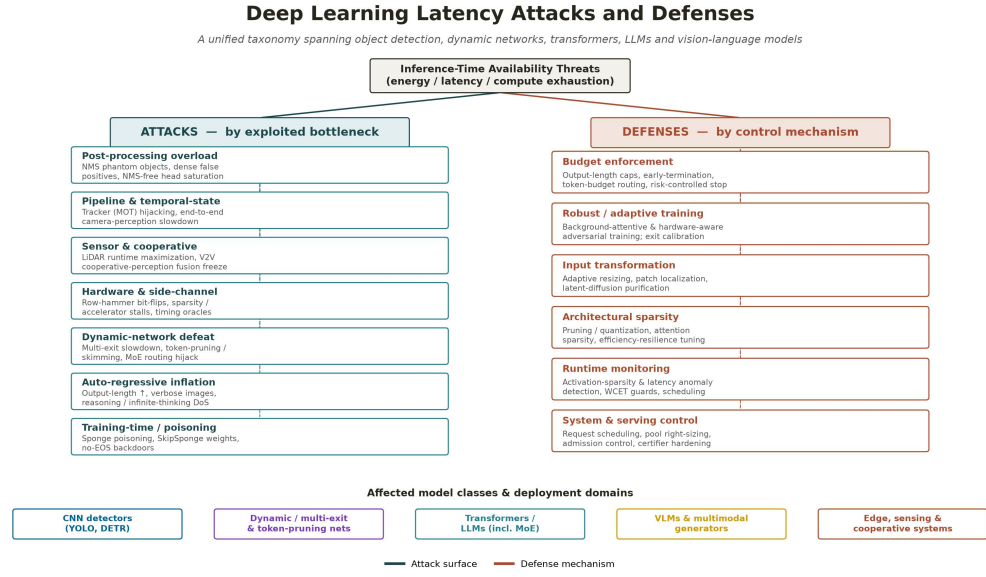


Fig. 1. Overview taxonomy of deep learning latency attacks and defenses. Attacks (left) are organized by the exploited computational bottleneck; defenses (right) by their control mechanism. The bottom band lists the model classes and deployment domains affected. The same content is provided as an editable slide.

2.2 Where latency comes from

A modern inference pipeline contains many cost centers, and latency attacks succeed by driving one or more toward its worst case:

- Data-dependent post-processing.** Operations whose cost depends on the *content*, not just the shape, of intermediate tensors. The canonical example is non-maximum suppression (NMS) in object detectors, whose cost grows with the number of candidate boxes [61].
- Activation sparsity.** Hardware accelerators (GPUs, TPUs, ASICs) exploit zero-valued activations to skip computation. Inputs that minimize sparsity force maximum multiply-accumulate work and energy [52, 62].
- Conditional computation.** Early-exit networks, token-pruning vision transformers, skimming language models, and mixture-of-experts routers all decide *at runtime* how much computation to perform [17, 56, 66]. Each decision point is an attack target.
- Auto-regressive generation.** Sequence models emit one token at a time until an end-of-sequence (EOS) signal. The number of tokens—and thus total cost—is determined by the model’s own outputs, which the attacker can steer [9, 23].
- Temporal and multi-agent state.** Pipelines that maintain state across frames (multi-object tracking) or aggregate across agents (cooperative perception) add stages whose cost depends on the number of tracked or fused entities [32, 70].
- Hardware and memory.** Memory access patterns, DRAM bandwidth, and even induced bit-flips can be manipulated to inflate latency below the algorithmic level [54, 63].

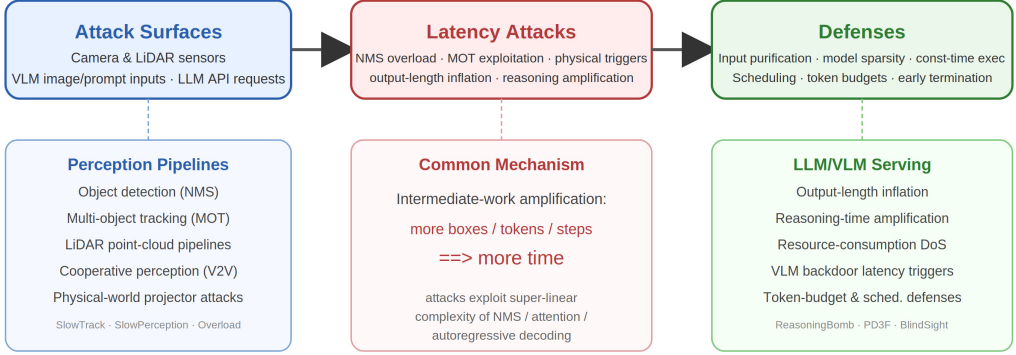


Fig. 2. The intermediate-work amplification principle unifies latency attacks across domains. Attack surfaces (left) feed attacks (center) that succeed by forcing more intermediate objects—boxes, tokens, or reasoning steps—through super-linear-complexity stages, which defenses (right) counter at different intervention layers. The bottom row contrasts the perception-pipeline and LLM/VLM-serving instantiations of the same mechanism.

2.3 The unifying principle: intermediate-work amplification

Beneath the diversity of targets and modalities lies a single mechanism, which we call *intermediate-work amplification*. In every latency attack, the adversary engineers a situation in which some internal boundary of the computation receives more objects, tokens, attention computations, or reasoning steps than a benign input of the same nominal class would generate. In object detection, those objects are phantom bounding boxes that flood NMS or multi-object tracking. In transformer inference, they are tokens that occupy the KV-cache and drive quadratic attention. In auto-regressive decoding, they are extra generated tokens; in reasoning models, extra chain-of-thought steps. The attack is effective precisely because the relevant algorithms—NMS, self-attention, autoregressive decoding, Hungarian matching—have at least *super-linear worst-case complexity* in the count of intermediate objects, so a modest increase in the count produces a disproportionate increase in cost. Figure 2 illustrates this shared structure across the perception and LLM/VLM-serving domains. This abstraction is more than descriptive: it implies that the natural cross-domain defense is a *work budget*—an explicit cap on the number of intermediate objects, tokens, or attention computations permitted per unit time, enforced at the boundary between stages (Section 11).

2.4 Related surveys

Adversarial-ML surveys overwhelmingly emphasize *evasion*—perturbations that cause misclassification—and treat the availability axis as peripheral [1, 26, 42]. Surveys of adversarial attacks on vision-language models catalog jailbreak, backdoor, and patch threats but largely omit output-length and resource-consumption attacks [13, 21, 29]. LLM-security surveys catalog serving-layer availability attacks but do not connect them to the perception-stack latency literature [78]. Earlier work by Kianpour and Wen [35] surveyed *timing attacks* on ML systems and laid conceptual groundwork, but predates the NMS-overload and LLM-DoS literature. The closest dedicated treatment, Brachemi Meftah et al. [4], covers energy-latency attacks on regular, adaptive, and decoder-based models but only shallowly addresses the LLM/VLM era. To our knowledge, this survey is the first to unify latency attacks and defenses across perception pipelines, dynamic networks, and LLM/VLM serving

Table 1. Positioning of this survey against prior surveys touching latency, timing, or availability attacks. ●: substantial coverage; ◐: partial/peripheral coverage; ○: not covered.

Survey	Year	AD percep.	General DL	LLM/VLM	Principal gap
Kianpour & Wen [35]	2019	○	◐	○	Predates modern latency/sponge and LLM-DoS literature
NIST AML taxonomy [68]	2023	○	◐	◐	Definitional only; availability is one small category
LLM-security surveys [1, 26, 42, 78]	2024–25	○	○	●	Serving-layer DoS only; no link to perception latency
VLM adversarial surveys [13, 21, 29]	2025–26	◐	○	◐	Jailbreak/backdoor/patch focus; omits output-length & resource attacks
Brachemi Meftah et al. [4]	2025	◐	●	◐	Shallow on LLM/VLM, reasoning, and MoE; no AD system-level synthesis
This survey	2026	●	●	●	— (unifies all three via intermediate-work amplification)

under a single threat model. Table 1 positions our work against prior surveys along three coverage axes—autonomous-driving (AD) perception, general deep learning, and the LLM/VLM era—and identifies the principal gap each leaves open.

2.5 A note on metrics

Reported results in this field are notoriously incomparable. Authors variously report multiplicative latency increase (×), absolute milliseconds, FLOP increase, energy in joules, generated-token count, frames per second, and downstream task outcomes (crash rate, collision rate). Throughout, we preserve each paper’s native metric and flag where direct comparison is unsound. Establishing common axes is itself an open problem (Section 14).

3 Threat Models

We classify latency attacks along three dimensions: the *attacker interface* (how the adversary touches the system), the *attacker knowledge* (white-, grey-, or black-box), and the *targeted stage*. Table 2 summarizes representative families.

3.1 Attacker interface

The interface ranges from purely digital (modifying the input tensor of a model under white-box control) to physical (projecting or printing adversarial patterns that a camera then captures) to systemic (injecting messages over a vehicle-to-vehicle channel, or submitting prompts to a public

Table 2. Threat-model families for latency attacks, with representative works.

Family	Interface	Targeted stage	Representative evidence
Digital input perturbation	Modify sensor input (image, point cloud, text)	Detector inference / NMS / decoder	Overload [6]; Phantom Sponges [61]; sponge examples [62]
Pipeline-level perturbation	Modify camera input for end-to-end effect	Full perception (detect + track)	SlowTrack [49]
Physical projector / patch	Project or print patterns into the scene	Camera perception via phantom objects	SlowPerception [50]; DetStorm [53]
Sensor runtime maximization	Perturb LiDAR point clouds	LiDAR detection runtime	SlowLiDAR [43]
Cooperative V2V injection	Inject perturbations via V2V messages	Cooperative-perception fusion	CP-FREEZER [70]
Auto-regressive inflation	Craft input prompt / image	Decoder generation length	NMTSloth [9]; Engorgio [14]; Verbose Images [23]
Dynamic-control defeat	Craft input to defeat conditional compute	Early-exit / pruning / routing	DeepSloth [28]; No-Skim [81]; Misrouter [18]
Training-time poisoning	Poison training data or weights	Persistent inference cost	Sponge poisoning [12]; Skip-Sponge [65]; DoS poisoning [24]
Hardware / side-channel	Induce faults or observe timing	NMS / memory / timing oracle	Bit-flip attack [63]; timing side-channel [54]

API). Physical realizability is the dividing line between laboratory demonstrations and deployable threats, and is a recurring theme in the autonomous-driving literature [50, 53].

3.2 Attacker knowledge

White-box attacks assume gradient access and dominate early work because the optimization is direct. Grey-box attacks assume architectural knowledge but not exact weights [15]. Black-box attacks, increasingly important for deployed services, rely on transfer from a surrogate, on query feedback (e.g., `usage.prompt_tokens`), or even on timing itself as an oracle [54, 84]. The existence of timing-only black-box attacks is significant: it means even an opaque service that returns no probabilities can leak enough information to be optimized against.

4 Latency Attacks on Object Detection and Perception

Object detection is the most thoroughly studied target, because its post-processing stage offers an unusually clean computational bottleneck and because autonomous driving supplies a high-stakes deployment context.

4.1 Non-maximum suppression as a bottleneck

NMS filters overlapping candidate boxes and has cost that grows with the number of candidates. *Phantom Sponges* [61] crafts a universal adversarial perturbation that adds many “phantom” boxes to overload NMS while *preserving* the original detections, so accuracy collapse does not betray

the attack. *Overload* [6] generalizes this by formulating latency maximization as an optimization with a spatial-attention mechanism, reporting that single-image inference time can be increased tenfold, and emphasizing that the attack is “NMS-agnostic.” *Beyond PhantomSponges* [60] modifies a box-area loss to lower intersection-over-union and exacerbate NMS load, reporting up to a 550% increase in NMS time on a YOLOv5-small configuration. *Daedalus* [71], although framed as an integrity attack, is a relevant NMS failure mode: it compresses box dimensions so NMS cannot filter dense false positives, and is demonstrated via printed posters. A complementary observation by Biton et al. [3] reframes variable-time inference as a security liability in itself: because NMS runtime is data-dependent, it *leaks* internal inference state, and this timing signal can be used to enhance black-box, decision-based attacks. This dual role—NMS as both a target of latency inflation and a side-channel oracle—motivates constant-time NMS as an explicit security requirement (Section 10).

4.2 Whole-pipeline and temporal-state attacks

Detector-only attacks often fail to change downstream behavior because multi-object tracking (MOT) absorbs detection noise; prior work shows that over 98% detection-attack success may be required to alter tracking outcomes [32]. This motivates pipeline-level threat models. *SlowTrack* [49] attacks camera-based perception end to end with a two-stage strategy and three new loss designs, reporting average latency 2.9× higher than prior approaches (e.g., raising SORT latency to 1,082 ms versus 411 ms for *Overload*) and, critically, connecting latency to safety: a vehicle crash rate of ~95% versus ~30% for baselines, evaluated on the Baidu Apollo stack and LGSVL simulator. Related temporal-integrity work—tracker hijacking [32, 48] and trajectory-aware attacks on MOT [30]—reinforces that stateful components can amplify or attenuate component-level perturbations.

4.3 Physical-world latency attacks

SlowPerception [50] introduces projector-based universal perturbations that create numerous phantom objects, inducing an average physical-world latency of 2.5 seconds and a 97% collision rate in an industry-grade system plus simulator. *DetStorm* [53] advances physical realizability, optimizing objects against multiple NMS variants and reporting a 506% average increase in detected objects and delays up to 8.1 seconds, while arguing that prior digital or large-patch attacks lack real-world applicability.

4.4 Beyond cameras: LiDAR and cooperative perception

SlowLiDAR [43] presents the first systematic availability attack on LiDAR detection, handling non-differentiable preprocessing via differentiable proxies and an execution-time-aware loss to inflate runtime across six popular pipelines while maintaining imperceptibility. *CP-FREEZER* [70] targets vehicular cooperative perception, injecting adversarial perturbations through V2V messages to maximize fusion delay; it resolves point-cloud non-differentiability and transmission asynchrony, increasing end-to-end latency by over 90×, pushing per-frame time beyond 3 seconds, and achieving 100% success on a real vehicle testbed. A simulation study further analyzes the system-level impact of perception inference-time attacks on full autonomous-vehicle stacks [7], and a real-time defense against object-based LiDAR attacks has begun to appear [82].

5 Sponge Examples and Energy-Latency Attacks on Neural Networks

The *sponge* lineage targets the general mechanism of activation sparsity rather than any specific post-processing stage, making it broadly applicable across architectures.

Table 3. Representative latency attacks on object detection and perception. Metrics are reported as in the source; “–” indicates not reported in a directly comparable form.

Method	Target stage	Key quantitative result	Physical?
Phantom	NMS (end-to-end	Universal perturbation overloads NMS	No
Sponges [61]	OD)	while preserving detections	
Overload [6]	OD inference (edge)	Single-image inference time up to 10×; NMS-agnostic	Edge-eval
Beyond Phantom-Sponges [60]	NMS runtime (YOLO)	Up to 550% NMS-time increase (YOLOv5-small)	No
SlowTrack [49]	Camera perception (detect+track)	Latency 2.9× vs. prior; crash rate ~95% vs. ~30%	Patch (prelim.)
SlowPerception [50]	NMS + MOT workload	Avg. 2.5 s physical latency; 97% collision rate	Yes (projector)
DetStorm [53]	Camera perception delay	506% more detections; up to 8.1 s delay	Yes
SlowLiDAR [43]	LiDAR detection runtime	Significant latency rise on six LiDAR pipelines	No
CP-FREEZER [70]	Cooperative fusion	> 90× latency; >3 s/frame; 100% on testbed	Yes (testbed)
Sponge	OD latency via NMS	Poisoned inputs prolong time; clean accuracy preserved	No
Backdoor [77]			
Bit-flip attack [63]	NMS via parameters	Up to 71.6 ms (20.4×) with 31 bit-flips	Partial

5.1 Inference-time sponge examples

Sponge examples [62] are the seminal work: adversarial inputs crafted to maximize energy and latency by minimizing the activation sparsity that accelerators exploit. On transformer and vision models, they report up to 30× increases in latency and energy in some settings, and a real-world demonstration raising a cloud translator’s response time from 1 ms to 6 s. A follow-up analysis shows that uniform/constant inputs act as “free” sponge examples and that the achievable energy increase is bounded by the architecture’s sparsity range, motivating activation-sparsity monitoring as a defense signal [52].

5.2 Sponge poisoning: moving the attack to training

Sponge poisoning [12] injects a small fraction of crafted samples so the trained model incurs elevated energy on *all* natural inputs thereafter—a far stealthier and more persistent variant. *SkipSponge* [65] operates directly on weights, poisoning residual (skip) connections with under 1% weight modification and evading fine-pruning and parameter-perturbation defenses. Sponge poisoning has been adapted to on-device DNNs [76], to multi-exit networks (where it both prevents early exits and raises per-layer cost) [31], to mobile apps [55], and to FPGA accelerators under differential privacy [2]. A sensing-AI study quantifies battery-drain risk in IoT settings and proposes pruning as a defense, reducing energy overhead from ~4× to ~1.3× [27].

5.3 Sparsity and spiking networks

Sparsity attacks [36] degrade energy/latency by minimizing activation sparsity and withstand activation thresholding and input quantization. Spiking neural networks, attractive for low-power edge perception, are also vulnerable: efficiency attacks raise spike activity by 1.7×–2.5× and energy by 1.4×–2.2× [37], and timestep-compressed methods reduce the latency of *generating* such attacks for real-time use [34].

6 Attacks on Dynamic and Adaptive Networks

Input-adaptive networks promise lower *average* cost; latency attacks revoke that promise by forcing worst-case control flow.

6.1 Multi-exit networks

DeepSloth [28] is the foundational slowdown attack on early-exit inference: per-input perturbations force the network to defer to its final, most expensive classifier, eliminating the efficiency advantage with 90–100% success and up to 5.2× cost inflation. Refined energy attacks combine energy maximization with confidence minimization and remain effective in grey-box settings [15]. In NLP, *SlowBERT* [80] attacks multi-exit BERT, and *Dynamic Transformers Provide a False Sense of Efficiency* [11] introduces a unified attack (SAME) that defeats early exits across BERT, RoBERTa, ALBERT, and DistilBERT with up to 5× slowdown. A foundational study demystifies *why* these models are vulnerable and proposes calibration-based defenses that recover roughly 60% of lost efficiency [67].

6.2 Token pruning, skimming, and mixture-of-experts

Skimming language models drop unimportant tokens; *No-Skim* [81] crafts inputs that prevent dropping, raising FLOPs by 2–3×. A unified framework treats early exit, token-pruning vision transformers, and MoE routing as instances of one “efficiency attack” class, demonstrating 2–8× overhead across all three [57]. Mixture-of-experts routers, central to modern sparse LLMs, are a fresh target: *RouteHijack* [79] concentrates tokens on expensive experts, and *Misrouter* [18] causes routing collapse or saturation in a black-box, query-only setting, both yielding 2–4× latency increases. *SPLAT* [39] revisits latency attacks on dynamic networks with a black-box, stealthy formulation for multi-exit models.

7 Latency Attacks on Sequence Generation: NMT, Captioning, TTS

Auto-regressive decoders introduce a uniquely powerful lever: the attacker can inflate the *number of generated tokens*, and total cost scales with it.

NMTSloth [9] is the first systematic study of efficiency robustness in neural machine translation, crafting source sentences that inflate decoder steps by 5–10× and end-to-end latency by 3–8× while leaving BLEU nearly unchanged so no quality monitor fires. *LLMEffiChecker* [19] extends this to general LLMs, finding them *more* vulnerable: a five-token perturbation can inflate generation length by 10–30×. *NICGSlowDown* [10] brings the paradigm to neural image captioning, forcing decoders to emit far longer captions via image perturbations, and establishes a widely reused efficiency-robustness benchmark. *TTSslow* [25] demonstrates that the threat reaches audio: adversarial text inflates autoregressive text-to-speech decoding by up to 5×. The common thread—and a recurring defensive blind spot—is that none of these attacks degrade output quality, so they evade quality-based detection entirely.

8 Latency and Resource-Consumption Attacks on LLMs and VLMs

The LLM/VLM era both inherits the auto-regressive lever and adds new ones: chain-of-thought reasoning, multimodal inputs, and shared serving infrastructure.

8.1 Output-length inflation and inference-cost DoS

Engorgio [14] optimizes prompts that coerce LLMs into babbling, inflating output token counts by 10–50× and transferring to black-box targets. *Crabs* [84] automates black-box LLM-DoS prompt generation, sustaining 10–30× output inflation even under provider-side length monitoring through

semantic camouflage. *Denial-of-Service Poisoning* [24] shows the threat can be embedded at training time: a triggered model emits endless output, never producing EOS, with 100% success at under 1% poisoning and survival through RLHF safety tuning.

8.2 Reasoning-model and “infinite-thinking” attacks

Inference-time compute scaling creates a new, high-leverage surface. *ReasoningBomb* [45] induces pathologically long reasoning traces in large reasoning models—averaging $\sim 18,759$ completion tokens versus ~ 300 for normal queries, a roughly $60\times$ cost multiplier—while remaining stealthy and bypassing input-, output-, and joint-detection baselines at rates of 99.8%, 98.7%, and 98.4%. *ThinkTrap* [40] exploits chain-of-thought mode in black-box APIs to induce near-infinite thinking loops, consuming $50\text{--}100\times$ more tokens than standard requests. These results argue strongly for budget enforcement *by construction* rather than detection.

8.3 Multimodal verbose attacks

Verbose Images [23] crafts adversarial images that make large VLMs (GPT-4V, LLaVA, InstructBLIP, MiniGPT-4) generate long, repetitive responses, inflating output by $10\text{--}30\times$ and latency by $7\text{--}30\times$ while preserving visual fidelity. *VLMInferSlow* [74] provides a service-oriented evaluation across twelve open and closed VLMs, reporting 85–100% success and $5\text{--}40\times$ latency increase. *Verbose-Text Induction* [47] co-optimizes adversarial images and prompts to exceed 10,000 output tokens from a single query, and *LingoLoop* [20] traps multimodal models in self-referential generation loops that fill the context window faster than legitimate output, defeating context-length caps. The trajectory from image captioning [10] through verbose images to multimodal loop attacks is a coherent and accelerating escalation. The threat has also reached *embodied* multimodal autonomy: a natural-reflection backdoor attack on driving VLMs [44] embeds reflection-pattern triggers in camera images that, when present, cause the VLM planner to generate abnormally long responses—demonstrating significantly increased inference latency under trigger with direct real-world safety implications, and uniting the output-length-inflation and training-time-backdoor threat classes in a single AD-relevant attack.

9 Training-Time, Hardware, and Side-Channel Attacks

Not all latency attacks act on the test-time input. *Training-time* variants (sponge poisoning [12], SkipSponge [65], sponge backdoors on detectors [77], and no-EOS DoS poisoning of LLMs [24]) embed persistent cost inflation that survives normal usage and is hard to remove without retraining. *Hardware* attacks operate below the algorithm: a row-hammer bit-flip approach targets NMS parameters via side channels, reporting up to 71.6 ms ($20.4\times$) latency with just 31 bit-flips [63], and FPGA sponge poisoning evades differential-privacy noise to inflate accelerator energy [2]. *Side-channel* attacks invert the relationship between timing and computation: a timing-based black-box method crafts adversarial examples using only processing time, exploiting the link between activated nodes and latency, and evades gradient-masking countermeasures [54]. This last result is conceptually important—it shows that latency is not only a target but also a leak.

10 Defenses

Defenses are nascent, often attack-family-specific, and pervaded by an efficiency–robustness tension. We organize them by *control mechanism*—the point of the compute pathway each one actually bounds—rather than by the attack they were designed against. Table 5 summarizes the landscape.

Table 4. Representative latency/energy attacks beyond object detection. Native metrics preserved.

Method	Domain / target	Key quantitative result
Sponge	General DNN	Up to 30× energy/latency; cloud translator 1 ms→6 s
Examples [62]	(vision, NLP)	
Sponge	Training-time, general	1–5% poison ⇒ persistent 2–10× energy
Poisoning [12]		
SkipSponge [65]	Weight poisoning	High energy overhead at <1% weight change; evades defenses
DeepSloth [28]	Multi-exit networks	90–100% force full inference; up to 5.2× cost
Dynamic Transformers (SAME) [11]	Multi-exit NLP	Up to 5× slowdown across BERT family
No-Skim [81]	Skimming LMs	2–3× FLOPs; output label unchanged
Misrouter [18]	MoE routing (black-box)	2–4× latency via routing collapse/saturation
NMTSloth [9]	Neural machine translation	5–10× decoder steps; 3–8× latency; BLEU stable
NICGSlowDown [10]	Image captioning	5–10× decoding steps; 3–7× latency
Engorgio [14]	LLM prompts	10–50× output tokens; black-box transfer
Crabs [84]	LLM-DoS (black-box)	10–30× output; evades length monitoring
ReasoningBomb [45]	Reasoning models	~60× tokens; >98% detection bypass
Verbose Images [23]	VLMs	10–30× output; 7–30× latency
VLMInferSlow [74]	VLM-as-a-service	85–100% success; 5–40× latency

10.1 Budget enforcement (bound the output / compute directly)

The most direct defense against auto-regressive inflation and reasoning DoS is to enforce a budget by construction. *PD3F* [83] is a two-stage, pluggable framework combining input-side dynamic request scheduling (via a “Resource Index”) with output-side “Adaptive End-Based Suppression” that terminates excessive generation early, improving access capacity by up to 500% under adversarial load across six models. *Token-budget-aware pool routing* [8] estimates each request’s token budget and routes it to a right-sized pool, avoiding worst-case provisioning that wastes 4–8× concurrency. *Conformal Thinking* [73] reframes the reasoning budget as risk control, stopping when the model is confident or when an instance is likely unsolvable, directly countering “runaway” reasoning. Because detection-based defenses are demonstrably fragile—ReasoningBomb bypasses them at >98% [45]—budget enforcement by construction is the more robust posture.

10.2 Robust and adaptive training

Hardware-aware adversarial training [72] builds background attention into the training pipeline using objectness loss as a proxy, restoring real-time throughput from 13 FPS to 43 FPS on a Jetson Orin NX under attack, with a better clean/robust trade-off. For multi-exit models, *calibration-based defenses* [67] (temperature scaling, robust exit calibration) recover roughly 60% of the efficiency lost to slowdown attacks.

10.3 Input transformation and purification

For NMS-exploitation attacks, *adaptive input resizing* [60] reduces NMS execution time by 90.18% on an attacked YOLOv5-large model. For physical patch/projector attacks, real-time defenses localize and remove adversarial patches via temporal consistency [51] or scale-adaptive DCT frequency analysis with inpainting [58], and integrated attack/defense studies report large mAP recovery and reduced detection time via texture-feature denoising [41]. For VLM visual encoders, *latent-diffusion purification* [38] projects inputs into a low-dimensional latent space to suppress off-manifold perturbations while preserving semantics.

10.4 Bounded and constant-time execution

A first-principles defense is to remove the data-dependence of cost entirely. Because many bottleneck operations have data-dependent runtime, rewriting them with a fixed worst-case bound eliminates both the latency-inflation lever and the timing side-channel. Biton et al. [3] explore constant-time NMS to close the timing-leakage channel they identify. More generally, capping the number of detections admitted to NMS, or the number of tokens a decoder may emit, converts an unbounded worst case into a bounded one—an instance of the *work budget* abstraction (Section 11). The cost is a potential accuracy ceiling in dense scenes, making the bound a tunable safety/utility parameter.

10.5 Architectural sparsity and efficiency reduction

Shrinking per-step compute reduces an attacker’s headroom. *Model pruning* [27] improves resilience to sponge poisoning (energy overhead $\sim 4\times \rightarrow \sim 1.3\times$) at an accuracy cost. *BlindSight* [64] exploits attention sparsity to cut VLM FLOPs by 32–41% with -2% to $+2\%$ accuracy. These reduce absolute latency impact but do not *bound* generation length, so they complement rather than replace budget enforcement.

10.6 Runtime monitoring and adaptive inference

Activation-sparsity monitoring offers a lightweight, architecture-agnostic anomaly signal for sponge inputs [52]. For early-exit systems, online scheduling treats the exit decision as a contextual bandit [33] (improving overall performance by up to 98.1% over the best static benchmark), and quorum-based ensemble stopping (SQUAD) [22] cuts inference latency by up to 70.6% while improving accuracy, by avoiding reliance on individual confidence scores that attackers manipulate.

10.7 System-level and serving control

At the serving layer, request scheduling, admission control, and pool right-sizing [8, 83] prevent operational bottlenecks from becoming attack amplifiers. A cautionary result: *certifiers*—formal robustness verifiers with abstain/fallback behavior—paradoxically create a new availability surface, since an adversary can systematically trigger abstention to deny service [46]. Any safety fallback is itself a target. Beyond admission control, hardware-level system defenses target the serving boundary directly: *Garrison* [75] is a GPU-accelerated inference system for adversarial ensemble defense that uses Multi-Instance GPUs with reinforcement-learning-based scheduling, improving adversarial robustness by up to 24.5% while accelerating ensemble inference by $6.6\times$ —a rare example of a defense that improves robustness *and* latency simultaneously. *PSML* [59] secures model-serving systems against model-extraction via fingerprinting and noise-based defenses, providing a complementary protection layer at the inference-serving boundary. The *time-traveling* defense [16] is an unusual input-side variant for the perception domain: it queries historical Street View imagery and uses majority-vote inference to resist adversarial traffic-sign manipulation,

Table 5. Defense families against latency/energy attacks, by control mechanism.

Family	Mechanism	Representative work & signal
Budget enforcement	Cap / terminate generation; route by budget; risk-control stop	PD3F (up to 500% capacity) [83]; token-budget routing [8]; Conformal Thinking [73]
Robust / adaptive training	Background-attentive & hardware-aware AT; exit calibration	Can't Slow Me Down (13→43 FPS) [72]; calibration (~60% recovery) [67]
Input transformation	Resize; patch localization; latent purification; temporal redundancy	Resizing (~90% NMS time) [60]; ADAV [51]; DCT [58]; LDP [38]; time-traveling (100%) [16]
Bounded execution	Constant-time / capped post-processing	Constant-time NMS [3]; detection / token caps
Architectural sparsity	Pruning / quantization; attention sparsity	Pruning ($4 \times \rightarrow 1.3 \times$) [27]; BlindSight (−32–41% FLOPs) [64]
Runtime monitoring	Sparsity/latency anomaly; online exit scheduling	Sparsity monitor [52]; DEE [33]; SQUAD (−70.6% latency) [22]
System / serving control	Scheduling; admission; GPU ensemble; certifier hardening	PD3F [83]; routing [8]; Garrison (+24.5% robust, 6.6×) [75]; PSML [59]; certifier caveat [46]

reporting 100% defense effectiveness in its evaluation, illustrating that temporal redundancy can substitute for per-frame robustness in geographically static scenes.

11 Cross-Domain Synthesis

Viewing the perception and LLM/VLM-serving literatures side by side yields three observations that neither community reaches alone.

Perception defenses do not transfer to serving, and vice versa. Hardware-aware adversarial training [72], input purification [38, 58], and the time-traveling majority vote [16] all assume the attack arrives through a *sensor input* as a pixel-level perturbation; they have no lever on prompt-injection DoS or output-length inflation, where the malicious content is carried in the semantics of a query rather than in pixel noise. Conversely, PD3F [83], token-budget routing [8], and conformal stopping [73] assume a serving stack with request queues and preemptible generation, which has no analog in a camera-to-detector pipeline. This bidirectional non-transfer is the central practical obstacle to securing a heterogeneous autonomy stack.

The intermediate-work amplification principle is universal, and so is its remedy. As argued in Section 2.3, every attack class derives its leverage from forcing a downstream stage to receive more inputs than it was provisioned for. The natural cross-domain abstraction is therefore a *work budget*: an explicit, enforced limit on the number of intermediate objects, tokens, or attention computations processed per unit time, applied at the boundary between stages. Constant-time NMS [3], detection caps, token budgets [8], and conformal reasoning bounds [73] are all instances of this single idea

expressed in different modalities. Recognizing them as one mechanism suggests that a general work-budget enforcement primitive, parameterized per stage, could provide uniform coverage.

No end-to-end defense exists. Every documented defense (Table 5) protects one layer and leaves others exposed. A VLM-equipped autonomous vehicle faces simultaneous latency threats at the camera-to-detector boundary (NMS overload, physical projectors), the tracking layer (SlowTrack-style temporal attacks), the LiDAR and cooperative-fusion stages, and the VLM planner’s decoder (verbose images, reflection backdoors). Composing defenses across these layers is technically possible but has not been evaluated end-to-end in any published work—a gap we return to below.

12 Evaluation Practices and Benchmarks

Progress is hampered by fragmented evaluation. Perception-attack papers use datasets such as MOT17, nuScenes, and KITTI, report wall-clock latency on specific GPUs (frequently an NVIDIA 2080 Ti or a Jetson device), and—in the strongest cases—report system-level outcomes on the Baidu Apollo stack and LGSVL simulator [49]. LLM/VLM-serving evaluations instead use vLLM-style serving benchmarks on models ranging up to Llama-3-70B, and report access capacity, token amplification, or detector-bypass rate [45, 83]. Three deficiencies recur. First, there is *no standardized latency threat model* analogous to the ℓ_p -ball used for evasion: papers leave attacker access, perturbation/query budget, and the target latency metric unspecified or mutually inconsistent. Second, there is *no agreed success metric*—wall-clock time, FPS, FLOPs, energy, crash rate, and access capacity are used interchangeably, defeating cross-paper comparison. Third, there is *no benchmark that jointly stresses perception and serving* under attack, which is precisely what an end-to-end AD-plus-VLM stack would require. We argue that joint latency/robustness protocols—measuring accuracy and timing *simultaneously* under adversarial load—are a prerequisite for trustworthy comparison, because efficiency and robustness interact non-monotonically (Section 14).

13 Emerging Surface: NMS-Free Detectors

A salient shift threatens to invalidate much of the object-detection attack literature: modern detectors increasingly remove NMS. YOLOv10 [69] and DETR-style models [5] replace NMS with one-to-one matching (Hungarian or dual-label assignment) and lightweight decoder heads, eliminating the quadratic post-processing blowup that Phantom Sponges and Overload exploit. This does not eliminate the threat—it relocates it. Plausible new bottlenecks include: the multi-scale feature-aggregation *neck* (PANet/BiFPN), where high-entropy features can be forced through all pyramid levels; the *dual-label assignment* head, where conflicting one-to-one and one-to-many decisions can stall the score comparator; the *task-aligned* top- k selection, where many near-identical scores maximize sorting work; *deformable attention*, where maximizing sampling-point spread induces non-coalesced memory access; and the *one-to-one matching* itself, where near-uniform assignment costs drive worst-case Hungarian iterations ($O(N^3)$). From a real-time-systems perspective, the matching-deadlock surface is especially compelling because it directly threatens worst-case execution-time (WCET) bounds—a well-defined failure mode in safety-critical scheduling. We highlight this as a concrete, under-explored direction: the persistence of latency threats across an architectural change that was *not* motivated by security.

14 Cross-Cutting Open Challenges

Standardized, system-level benchmarks. Most attacks report per-component timing in incompatible units; few connect to downstream outcomes. SlowTrack’s crash-rate evaluation [49] and CP-FREEZER’s testbed [70] are exceptions. The field needs benchmarks that report *both* timing and safety/serviceability impact on common axes.

Physical realizability. Projector attacks [50, 53] demonstrate multi-second physical delays but require active infrastructure; SlowTrack’s physical transfer remains preliminary [49]. The gap between strong digital results and robust physical deployment is wide.

Inference-time compute as a new surface. Reasoning models and MoE routers [18, 40, 45] created entirely new attack classes within a single year. As architectures continue to make compute input-dependent, each such mechanism is a candidate target, and defenses lag behind by construction.

Timing as an oracle. Black-box timing attacks [54] show that even opaque deployments leak optimization signal, complicating any defense that assumes attacker ignorance.

The efficiency–robustness tension. Every architectural defense—pruning, sparsity, early exit—trades average-case efficiency against worst-case robustness, and the trade-off is non-monotone [27, 65]. Defenses that bound worst-case cost by construction (budget enforcement, WCET guards) are more promising than those that merely reduce average cost.

Detection fragility. High detection-bypass rates [45] argue that anomaly detection alone is insufficient; enforceable budgets and scheduling must complement it.

End-to-end stack evaluation. A modern autonomy stack with a VLM planner is simultaneously vulnerable at the camera sensor, at NMS, at MOT, at LiDAR/cooperative fusion, and at the LLM decoder, yet no published work evaluates all of these jointly. Building a simulation environment that injects attack stimuli at each layer and measures *system-level* outcomes—collision rate, mission completion, service availability—is, in our view, the most practically important open engineering problem in this space, and the natural home for the work-budget primitive proposed in Section 11.

Defense non-transfer. Because perception-side and serving-side defenses operate on incompatible attack surfaces (Section 11), a heterogeneous stack must currently compose multiple narrowly scoped defenses. Whether a single unifying primitive—work-budget enforcement at every stage boundary—can subsume them without unacceptable accuracy cost is an open question.

15 Conclusion

Latency attacks extend the security of deep learning beyond correctness to timeliness and resource availability. What began as NMS overload on a single detector has matured into a cross-cutting threat spanning physical autonomous-driving perception, sponge attacks on accelerators, slowdown attacks on every form of conditional computation, and a rapidly expanding frontier of output-length, verbose-image, and reasoning denial-of-service attacks on LLMs and VLMs. The unifying lesson is that *any* mechanism which makes inference cost input-dependent—post-processing, sparsity, early exit, routing, auto-regressive generation, inference-time reasoning—becomes an attack surface the moment efficiency is advertised. Defenses are advancing, with hardware-aware training, input purification, and especially budget-enforcement-by-construction showing the most promise, but they remain fragmented and bound by a persistent efficiency–robustness tension. Progress will depend on standardized benchmarks that report timing alongside downstream impact, on threat models that take physical and serving realism seriously, and on defenses that control worst-case computational paths rather than merely lowering the average. As deep learning continues to trade fixed cost for input-adaptive efficiency, the availability axis will only grow in importance.

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A Catalog of Latency, Energy, and Timing Attacks and Defenses

This appendix provides a structured, citable catalog of the works surveyed, organized by attack stage (inference vs. training) and by defenses. Each entry records the target architecture, application domain, attacker setting, and links to the paper and released code where available. A continuously updated, searchable web version of these tables is maintained as a companion repository at <https://github.com/guzonghua/awesome-latency-attacks>. Tables 6–8 mirror that resource.

Table 6. Inference-stage latency, energy, and timing attacks on deep learning.  links to the paper;  links to released code (X = none located).

Attack	Venue (Year)	Target	Domain	Setting	Paper	Code
Daedalus	arXiv (2019)	Object detection (NMS)	CV / AD	White box (physical)		
Sparsity Attacks	IEEE TCAD (2020)	CNNs	CV	White box		X
Tracker Hijacking	ICLR (2020)	Multi-object tracking	AD	White box		X
Sponge Examples	EuroS&P (2021)	Transformers + CNNs	CV & NLP	Black & white box		
DeepSloth	ICLR (2021)	Multi-exit CNNs	CV	White box		
Timing Side-Channel AE	IEICE Trans. (2021)	DNN classifiers (timing oracle)	CV	Black box (timing)		X
SpikeAttack	ACM/IEEE DAC (2022)	Spiking neural networks	CV	White box		X
NMTSloth	ESEC/FSE (2022)	Decoder-based NMT	NLP	White box		
LLMEffiChecker	ACM TOSEM (2022)	LLMs	NLP	Black & white box		
NICGSlowDown	CVPR (2022)	Decoder-based image captioning	CV	White box		
SAME	ACL (2023)	Multi-exit transformers	NLP	White box		
SlowBERT	ACL Findings (2023)	Multi-exit BERT	NLP	White box		X
No-Skim	arXiv (2023)	Skimming language models	NLP	Black & white box		X
Phantom Sponges	WACV (2023)	Object detection (NMS)	CV / AD	White box		
SlowLiDAR	CVPR (2023)	3D LiDAR detection	CV / AD	White box		
WIP Tracker Attack	VehicleSec (2023)	Multi-object tracking	AD	White box (patch)		X
Variable-Time Inference	AISeC@CCS (2023)	Object detection (NMS timing)	CV	Black box (timing)		X
Overload	CVPR (2024)	Object detection (edge)	CV / AD	White box		X
Beyond Phantom-Sponges	ACM WiseML (2024)	Object detection (NMS)	CV / AD	White box		X
SlowTrack	AAAI (2024)	Camera perception (detect+track)	AD	White box		
SlowPerception	arXiv (2024)	Camera perception (NMS+MOT)	AD	White box (physical, projector)		X
TPA (Time-aware)	Pattern Recognition (2024)	Detection + segmentation	AD	White box		X
Steal Now Attack Later	arXiv (2024)	Object detection	CV	Black box		X
Energy Attack (multi-exit)	Info. & Software Tech. (2024)	Adaptive multi-exit networks	CV	Grey box		X
Slowdown Causes (SaTML)	IEEE SaTML (2024)	Language models	NLP	White box		X
Engorgio	arXiv (2024)	LLMs (output inflation)	NLP	White box + transfer		X
Verbose Images	ICLR (2024)	Large VLMs	CV+NLP	White box		
Uniform Inputs	IEEE SPW (2024)	CNNs (sparsity)	CV	Black box		
DetStorm	IEEE S&P (2025)	Camera perception	AD	White box (physical)		X

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Table 6. Inference-stage latency, energy, and timing attacks on deep learning.  links to the paper;  links to released code (X = none located). (continued)

Attack	Venue (Year)	Target	Domain	Setting	Paper	Code
Inference-Time Impact Analysis	arXiv (2025)	Full perception (sim.)	AD	Simulation		X
DDLs Efficiency Attacks	arXiv (2025)	Early-exit / token-pruning / MoE	CV & NLP	White & black box		X
TTSlow	IEEE TASLP (2025)	Auto-regressive TTS	Speech	White box		X
Crabs	ACL Findings (2025)	LLMs (DoS)	NLP	Black box		X
VLMInferSlow	ACL (2025)	VLMs-as-a-service	CV+NLP	Black box		X
Verbose-Text Induction	arXiv (2025)	VLMs	CV+NLP	White box		X
LingoLoop	arXiv (2025)	Multimodal LLMs	CV+NLP	White box		X
Bit-Flip NMS Attack	WACV (2025)	Object detection (parameters)	CV / AD	Hardware (Rowhammer)		X
Timestep-Compressed Attack	AAAI (2025)	Spiking neural networks	CV	White box		X
CP-FREEZER	AAAI (2026)	Cooperative perception (V2V)	AD	White box (testbed)		X
Trajectory-Aware Attack	IEEE TMM (2026)	Multi-object tracking	AD	White box		X
SPLAT	IEEE TCAD (2026)	Multi-exit dynamic networks	CV	Black box		X
RouteHijack	arXiv (2026)	Mixture-of-experts LLMs	NLP	White box		X
Misrouter	arXiv (2026)	Mixture-of-experts LLMs	NLP	Black box (input-only)		X
ReasoningBomb	ACM CCS (2026)	Large reasoning models	NLP	Black box		X
ThinkTrap	NDSS (2026)	Reasoning LLM APIs	NLP	Black box		X

Table 7. Training-stage (poisoning / backdoor / weight) latency and energy attacks.

Attack	Venue (Year)	Target	Domain	Setting	Paper	Code
Sponge Poisoning	arXiv / AsiaCCS (2022)	CNNs	CV	Partial control		
On-Device Sponge Poisoning	ACM SecTL (2023)	On-device DNNs	CV	Partial control		X
Mobile-App Sponge	ACM HotMobile (2023)	Mobile ML models	CV	Partial control		X
SkipSponge	arXiv (2024)	CNNs, GANs (weights)	CV	Full control		X
Huang et al. (multi-exit)	IEEE Access (2024)	Multi-exit CNNs	CV	Full control		X
Sponge Backdoor (OD)	IJCNN (2024)	Object detection (NMS)	CV / AD	Backdoor		X
DoS Poisoning (LLM)	arXiv (2024)	LLMs (no-EOS)	NLP	Backdoor / poisoning		X
Sensing-AI Sponge	IEEE GLOBECOM (2025)	Sensing DNNs (IoT)	Sensing	Partial control		X

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Table 7. Training-stage (poisoning / backdoor / weight) latency and energy attacks. (continued)

Attack	Venue (Year)	Target	Domain	Setting	Paper	Code
EvoWeight (FPGA)	IEEE HOST (2025)	FPGA DNN accelerators	CV	Full control		
Reflection Backdoor (VLM-AD)	arXiv (2025)	Driving VLM planner	AD	Backdoor (physical trigger)		

Table 8. Defenses against latency, energy, and timing attacks, by control mechanism.

Defense	Venue (Year)	Target	Mechanism	Domain	Paper	Code
DEE Scheduling	ACM CIKM (2021)	Early-exit networks	Runtime scheduling	CV		
Certifier Caveat	arXiv (2021)	Certified classifiers	Defense pitfall	CV		
Constant-Time NMS	AISeC@CCS (2023)	Object detection (NMS)	Bounded execution	CV		
PSML	arXiv (2023)	Inference serving systems	System / serving control	ML serving		
Adaptive Resizing	ACM WiseML (2024)	Object detection (NMS)	Input transformation	AD		
ADAV Patch Defense	arXiv (2024)	Object detection	Input transformation	AD		
Securing AV Perception	IEEE TIV (2024)	AV visual perception	Input transformation	AD		
Calibration Defense	IEEE SaTML (2024)	Multi-exit models	Robust training	NLP		
Sparsity Monitor	IEEE SPW (2024)	CNNs	Runtime monitoring	CV		
Garrison	ACM/IEEE DAC (2024)	Ensemble inference (GPU)	System / serving control	CV		
Time-Traveling Defense	arXiv (2024)	Traffic-sign classifiers	Temporal redundancy	AD		
Can't Slow Me Down	CVPR (2025)	Edge object detectors	Robust/adaptive training	AD		
DCT Patch Elimination	IEEE RCAR (2025)	Object detection	Input transformation	AD		
Real-Time LiDAR Defense	ACM CCS (2025)	LiDAR detection	Runtime monitoring	AD		
LDP Purification	IEEE TrustCom (2025)	VLM visual encoders	Input purification	CV+NLP		
Pruning Defense	IEEE GLOBECOM (2025)	Sensing DNNs	Architectural sparsity	Sensing		
BlindSight	arXiv (2025)	VLMs	Architectural sparsity	CV+NLP		
PD3F	EMNLP (2025)	LLM serving	Budget enforcement	NLP		
SQUAD	arXiv (2026)	Early-exit ensembles	Runtime scheduling	CV		
Token-Budget Routing	arXiv (2026)	LLM serving	Budget enforcement	NLP		
Conformal Thinking	arXiv (2026)	Reasoning models	Budget enforcement	NLP		